28 The distance from the Earth to the Sun is 1.496×10^{11} metres.
The speed of light is 3×10^{8} metres per second.

(a) Show that, correct to 3 significant figures, light will take 0.139 hours to travel from the Sun to the Earth.

Time = distance ÷ speed

$= (1.496 \times 10^{11}) \div (3 \times 10^{8}) = 498.666...$ seconds.

1 mark
Distance ÷ speed

Hours = 498.666... ÷ 3600 = 0.1385185185...

1 mark
Convert to hours

= 0.139 hours to3 significant figures, as required.

1 mark
Shown0.139h

(3)

1 googol is 1×10^{100}

Danesh says,

"When I multiply 1.496×10^{11} by 6.68×10^{9} I get nearly 1 googol because $1.496 \times 10^{11} \times 6.68 \times 10^{9} = 9.99 \times 10^{99}$ "

(b) Is Danesh correct?
Give a reason for your answer.

No. Multiply powers of10 by adding their indices:

$10^{11} \times 10^{9} = 10^{20}$, not$10^{99}$.

Product = $1.496 \times 6.68 \times 10^{20} = 9.99328 \times 10^{20}$.

1 mark
Add indices11+9=20; do not multiply them

(1)

(Total for Question 28 is 4 marks)